

Evolution and Unified Performance Evaluation of Clustering Techniques in Heterogeneous Wireless Sensor Networks

Ashish Sharma*, Yogesh Sharma* Vansh Tomar[†], Satvik Rastogi[†], Ansh Rai[†]

*Assistant Professor, Department of Computer Science and Engineering

Maharaja Agrasen Institute of Technology, Delhi, India

ashish@mait.ac.in, yogeshsharma@mait.ac.in

[†]Students, Department of Computer Science and Engineering

Maharaja Agrasen Institute of Technology, Delhi, India

vanshwar@gmail.com, satvikrastogi777@gmail.com, raiansh230405@gmail.com

Abstract—Clustering protocols for wireless sensor networks span three methodological generations: probabilistic heuristics, explicit optimization, and learned selection functions. They are usually compared by reading their published tables side by side, which does not work, because each was measured against a different simulator, field size, node count, energy budget and channel. We rebuild the comparison. Nine protocols plus a direct-transmission baseline run on one engine that owns energy accounting, packet sizing, fusion, the channel and the counting of delivered readings; a protocol returns only a cluster structure, and a static audit confirms that no implementation touches the energy or liveness arrays. Trials are paired, so every protocol sees the same topology, shadowing and sensed-value stream, and differences are tested with sign-flip permutation, paired bootstrap intervals and Holm correction. Running the study at both endpoints of the one parameter the source specifications leave open, transmit power, retracts one of our own comparisons: the first-death advantage of the type-2 fuzzy protocol and the deep Q-network over LEACH is 417 and 426 rounds under packet loss and falls to 17 and 22 rounds, no longer significant, once loss is removed. Their advantage in overall survival and in readings delivered survives both channels. A 1,350-run sweep over three node counts and three field areas finds the same boundary from the other side: distance to the sink, not node density, sets the regime, and in a 50×50 m field clustering is worse than no clustering at all. Mechanisms are reported by measurement rather than argument, and the modeling choices that bound these results are disclosed in full.

Index Terms—Wireless sensor networks, clustering protocols, network lifetime, energy efficiency, reinforcement learning, graph neural networks, type-2 fuzzy logic, cluster head selection, reproducibility.

I. INTRODUCTION

A wireless sensor node spends most of its energy on its radio, and in nearly all deployments its battery cannot be replaced [1]. Lifetime is decided by how the network organizes its transmissions rather than by any property of the sensing hardware. Clustering is the standard response: nodes elect a subset of themselves as cluster heads, members send short-range packets to their head, and the head fuses them into one packet and pays the single expensive transmission to the base station. Because that role is expensive it has to rotate, and

the question every clustering protocol answers is which nodes should carry it this round.

Twenty-five years of answers fall into three generations. The first is heuristic: LEACH rotates the role by a probabilistic threshold with an epoch constraint [8], PEGASIS abandons clusters for a greedy chain in which each node talks only to its nearest neighbor [10], and TEEN and APTEEN leave the topology alone and suppress transmissions below a sensed-value threshold [11], [12], with later heuristics such as HEED refining the election rule without changing its character [2], [13]. The second generation replaces the heuristic with an explicit objective, solving head selection as a multi-objective problem with NSGA-II [16] or encoding it in a type-2 fuzzy rule base over residual energy, distance and local density [19], [20]. The third learns the selection function instead of specifying it [3]: a self-organizing map places heads by unsupervised vector quantization [25], a deep Q-network treats rotation as a sequential decision problem [27], [28], and a graph convolutional network scores candidates from the adjacency structure [29], [30].

The generations are usually compared by reading their published tables side by side, and that comparison does not hold. Each protocol was proposed against a different simulator, field size, node count, packet length and energy budget [4], [5], [6], almost always against LEACH alone and almost always on a lossless channel. More consequentially, the assumptions that decide the outcome are rarely stated. Whether a centralized protocol is charged for the uplink that reports node state can be worth a quarter of the energy budget. Whether fusion is lossless in bits determines how much a chain protocol gains over a clustered one. Whether lifetime is reported at first, half or last death can reverse a ranking outright, and the author chooses which to report. A margin obtained under one set of these choices is not evidence about the algorithm; it is partly evidence about the harness.

This paper measures the three generations in one harness. Nine protocols plus a direct-transmission baseline run over 30 paired trials in each of two channel configurations, for

600 runs, and over a further 1,350 runs in a node-count by field-area sweep. Run index i seeds the topology, the per-link shadowing and the sensed-value stream identically for every protocol, so a comparison at a given index is genuinely paired rather than an average over unrelated worlds. Fairness is enforced structurally: the engine owns energy accounting through a single charging function, packet sizing, the aggregation charge, the channel and its ARQ retries, and the counting of delivered readings. A protocol returns only a cluster structure and a route shape, and a static audit verifies that no implementation writes to the energy or liveness arrays. Everything runs on the same numerical substrate, hand-written NumPy [46] including the backpropagation for both learned models, because per-round setup time is a reported metric. No protocol is tuned to reproduce any published number; disagreement with the source tables is expected and is not treated as error.

One assumption favors part of the field and we state it up front. Centralized protocols are not charged for the uplink carrying node state to the sink, because that state is treated as piggybacked on data traffic the network already sends. This is standard in the centralized-clustering literature but it is still a choice, and Section VI measures it: LEACH spends 12.73% of its consumed energy on control traffic against 2.33% to 3.95% for the centralized five, so every cross-class comparison carries a subsidy of nine to ten percentage points of the budget.

Four results anchor what follows and Section VII restates them with their numbers. Transmit power is not a background parameter but a determinant of the ranking, and bracketing it retracts one of our own comparisons. The mechanism behind that retraction turns out to be the far tail of the head-to-sink hop rather than its mean, which is where we first looked. Clustering redistributes lifetime rather than extending it, so no single death point summarizes it. And the value of clustering is a property of an operating regime: when the sink is close enough that no link reaches the packet-error waterfall, the no-clustering baseline outlives LEACH.

Section II describes the system model, Section III the protocols, Section IV the experimental design, Section V the results, Section VI the threats to validity, and Section VII concludes.

II. SYSTEM MODEL

A. Network and Energy

One hundred nodes are placed uniformly at random in a 100×100 m field, with the base station outside the field at (50, 150), so node-to-sink distances span 50 to 158 m. Every node begins with $E_0 = 1$ J, is stationary, and fails only by depletion. The configuration carries fields for two-level and three-level heterogeneity [14], [15], disabled throughout, so every result here is for a homogeneous population. Table I lists every parameter; nothing in it is protocol-specific, and no constant is defined outside the single frozen configuration object.

A run proceeds in rounds until the last node dies or a cap of 7,000 rounds is reached. Energy is a linear reservoir with a

TABLE I
SIMULATION PARAMETERS. EVERY VALUE IS READ FROM THE SINGLE FROZEN CONFIGURATION OBJECT USED BY ALL TEN PROTOCOLS; NOTHING IS PROTOCOL-SPECIFIC.

Parameter	Value
<i>Topology and energy</i>	
Nodes N	100
Field	100×100 m
Base station	(50, 150) m
Initial energy E_0	1.0 J per node
Rounds (cap)	7,000
Runs per configuration	30
<i>Packets</i>	
Data packet	4,000 bits
Control packet	200 bits
<i>Radio (first-order model)</i>	
E_{elec}	50 nJ/bit
ε_{fs}	10 pJ/bit/m ²
ε_{mp}	0.0013 pJ/bit/m ⁴
E_{DA} (fusion)	5 nJ/bit
Crossover d_0	87.7 m
<i>Channel</i>	
Transmit power P_t	12 dBm
Path loss at 1 m	40 dB
Path-loss exponent n	3.0
Shadowing σ	4 dB
Noise floor	-105 dBm
ARQ retransmissions	2
<i>Sensed data (Ornstein–Uhlenbeck)</i>	
Mean μ , reversion θ	50, 0.05
Step σ	2.0
TEEN/APTEEN H_T , S_T , T_C	55, 1.0, 50 rounds
<i>Protocol-agnostic</i>	
Head fraction p	0.05
Density radius	25 m
NSGA-II population, generations	40, 30
Re-clustering interval	5 rounds
<i>Timing</i>	
Bit rate	250 kbps
Slot / guard / processing	20 / 4 / 1 ms

single charging function: a node asked to spend more than it holds spends what remains, dies, and loses the packet in flight. Refusing the transmission instead would strand the remainder and break the conservation identity

$$\sum_i (E_0(i) - E_{\text{res}}(i)) = E_{\text{tx}} + E_{\text{rx}} + E_{\text{ctrl}} + E_{\text{agg}} + E_{\text{retry}}, \quad (1)$$

asserted at the end of every run to 10^{-9} J and the primary correctness check on the simulator.

B. Radio Energy

We use the first-order model of Heinzelman et al. [8], [9]. Transmitting k bits over distance d costs

$$E_{\text{tx}}(k, d) = \begin{cases} kE_{\text{elec}} + k\varepsilon_{\text{fs}}d^2, & d < d_0 \\ kE_{\text{elec}} + k\varepsilon_{\text{mp}}d^4, & d \geq d_0 \end{cases} \quad (2)$$

with $E_{\text{rx}}(k) = kE_{\text{elec}}$ and a fusion charge of E_{DA} per bit received. The crossover $d_0 = \sqrt{\varepsilon_{\text{fs}}/\varepsilon_{\text{mp}}} = 87.7$ m sits inside the range of node-to-sink distances, which makes head

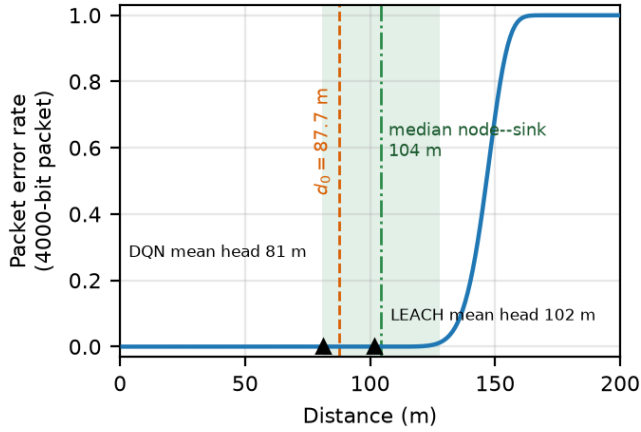


Fig. 1. Packet error rate against distance for a 4,000-bit packet, with the crossover $d_0 = 87.7$ m and the measured band of node-to-sink distances marked. The curve is a waterfall, not a gradient: effectively zero out to about 120 m, saturated by 158 m. Where a protocol places the far tail of its head-to-sink hop relative to that step decides what it pays in retransmissions, which is why transmit power is an experimental factor in Section IV rather than a constant.

placement relative to d_0 a live variable: a head at 100 m pays the fourth-power branch and one at 80 m does not. In the seed-0 topology 36 of the 100 nodes fall inside d_0 , against an expected 32.7% over the placement distribution.

The model is minimal and its omissions matter. There is no idle listening, no sleep-wake transition energy, no radio startup and no processor energy. Idle listening often dominates real budgets [39], [40], and including it would compress the differences between all nine protocols, so this bounds every energy figure that follows.

C. Channel

Packet loss is derived from a link budget rather than assigned as a constant. Path loss follows a log-distance law with per-link shadowing [34],

$$PL(d) = PL_{1m} + 10n \log_{10} d + X_\sigma, \quad X_\sigma \sim \mathcal{N}(0, \sigma^2), \quad (3)$$

giving a receive power of $P_t - PL(d)$ and an SNR against a -105 dBm noise floor. Bit errors follow the closed-form non-coherent BFSK expression [35] $BER = \frac{1}{2}e^{-\gamma/2}$, and $PER = 1 - (1 - BER)^k$, evaluated as $-\expm1(k \ln(1 - BER))$ for numerical stability.

Fig. 1 plots the result, and it is a waterfall rather than a gradient. Without shadowing, PER is 2.6×10^{-8} at 100 m, 1.0×10^{-3} at 120 m, 0.19 at 140 m and 0.97 at 158 m, so the transition from usable to useless spans about 3.6 dB of SNR and 40 m. Shadowing at $\sigma = 4$ dB smears that step into a distribution over links and is the source of the genuine per-topology variance in delivery ratio; it is drawn once per topology and held fixed, because the nodes are static. Failed packets are retried up to twice, each attempt paying full transmit and receive energy, which is what makes a lossy channel cost joules rather than only packets. Retry energy is

charged to a separate category so it can be measured directly rather than inferred.

D. Control-Traffic Accounting

Control traffic is charged by the engine according to the protocol's declared class, not by the protocol itself, and both endpoints pay. Distributed protocols pay a full handshake every round: an advertisement priced at the distance to the farthest joiner with a receive charge on every living non-head, a join request per member, and a TDMA schedule broadcast every member receives. The chain protocol passes a leader token hop by hop instead, plus a sink broadcast when the chain is rebuilt. Centralized protocols skip the advertisement and join-request steps, since the sink has already told each node its assignment, but do still pay the per-round schedule broadcast under identical pricing. Their uplink node state is treated as piggybacked on data traffic and charged nothing. The subsidy is therefore narrower than "centralized protocols pay almost no control traffic": it is specifically the advertisement, the join request and the uplink report, which is why the measured control-energy share of the centralized five is 2.33% to 3.95% rather than zero.

That choice is the largest modeling decision in the paper. Section VI-A gives its magnitude and what it costs every cross-class comparison.

E. Sensed Data, Latency and Invariants

Each node's reading follows a discrete Ornstein-Uhlenbeck process [41] stationary at approximately $\mathcal{N}(50, 6.4^2)$; a pure random walk was rejected because over 7,000 rounds it pegs against any clipping bound, which would make the reactive protocols' behavior an artifact of the clip. TEEN and APTEEN's thresholds are set against that stationary distribution, hard at $+0.78\sigma$ and soft at 0.16σ , and were not adjusted after observing any result. The full sensed stream is generated once per topology, so every protocol reads byte-identical values.

Latency is analytic, $L = s T_{\text{slot}} + h (T_{\text{tx}} + T_{\text{proc}})$, with s the sender's TDMA slot index and h its hop depth. A head's own packet carries a slot index equal to its member count, which is what makes cluster latency scale with cluster size. There is no queueing or contention delay, so absolute figures are lower bounds and only the ordering should be read.

Three rules cannot be bypassed by a protocol implementation. Transmissions execute sorted by hop depth and then node index, because plain index ordering breaks causality for roughly half of all head-member pairings on a random topology and silently corrupts fusion totals. A node that receives data emits at most one outgoing packet and pays the fusion charge on every bit received. And the count of original readings in a delivered packet is computed by the engine, not declared by the protocol, since an earlier version that let protocols declare it would have let a protocol inflate its own delivery figures.

TABLE II

THE TEN CONFIGURATIONS COMPARED. “CONTROL MODEL” DETERMINES WHICH SETUP MESSAGES ARE CHARGED; DISTRIBUTED PROTOCOLS PAY ADVERTISEMENT, JOIN AND SCHEDULE; THE CHAIN PROTOCOL PAYS A HOP-BY-HOP TOKEN; CENTRALIZED PROTOCOLS PAY ONLY THE DOWNLINK SCHEDULE, THEIR UPLINK STATE BEING PIGGYBACKED ON DATA TRAFFIC (SECTION VI).

Protocol	Gen.	Head-selection rule	Control model	Re-cluster	Offline train
Direct (no clustering)	–	None	None	–	No
LEACH	G1	Randomized rotation	Distributed	Every round	No
PEGASIS	G1	Greedy chain	Chain	On death	No
TEEN	G1	LEACH + hard/soft threshold	Distributed	Every round	No
APTEEN	G1	TEEN + count-down timer	Distributed	Every round	No
NSGA-II	G2	Multi-objective search (3 obj.)	Centralized	Every 5 rounds	No
Fuzzy T2	G2	Interval type-2 rule base (27 rules)	Centralized	Every round	No
SOM	G3	Self-organizing map	Centralized	Every round	Yes
DQN	G3	Learned Q -value ranking	Centralized	Every round	Yes (frozen)
GCN	G3	Learned graph scoring	Centralized	Every round	Yes (frozen)

III. PROTOCOLS UNDER COMPARISON

Nine protocols are implemented, spanning three generations, plus a direct-transmission baseline in which every node sends its own reading straight to the sink. Table II sets out the ten configurations against class, control-traffic model, re-clustering cadence and pre-training.

All nine share one structural contract. Each targets the same nominal cluster count $k = \max(1, \text{round}(p \cdot N_{\text{alive}}))$, recomputed each round so k decays as the network dies, with membership always nearest-head by Euclidean distance. What each protocol returns is a set of head identities and a membership map, nothing else; energy, packet sizing, fusion charges, the channel and reading accounting belong to the engine (Section II-E). Realized head counts are not forced to match the target, because clamping LEACH’s stochastic election would be implementing a different algorithm.

A. Generation One: Heuristics

LEACH [8] implements the original election threshold unmodified: a node that has not yet been a head in the current epoch self-elects with probability $T(n) = p/(1 - p(r \bmod 1/p))$, and the served set clears every $1/p = 20$ rounds. The epoch mechanism turns out to be LEACH’s real strength, because it guarantees each surviving node serves exactly once per epoch, close to optimal for first death specifically, and no other protocol here has an equivalent guarantee.

PEGASIS [10] abandons clusters for a greedy chain built from the node farthest from the sink, rebuilt only when a death invalidates it, with the leader rotating round-robin over chain positions. Data flows inward from both ends and fuses at every hop, so each non-leader emits one packet and the whole network’s readings reach the sink in a single transmission. PEGASIS is the sole multi-hop protocol here and the sole beneficiary of ideal fusion at full scale.

TEEN [11] subclasses LEACH and reuses its clustering unchanged. This is the correct reading of the protocol rather than a shortcut: TEEN’s contribution is entirely the reporting rule, and holding clustering fixed makes the comparison controlled. A node transmits only when its reading exceeds a hard threshold and has moved by a soft threshold since its last transmission. APTEEN [12] adds the periodic half, so a

node that never crosses the hard threshold still reports every 50 rounds. Only the reporting-frequency mechanism is modeled; APTEEN’s query facilities are out of scope.

B. Generation Two: Explicit Optimization

NSGA-II [16] poses head selection as a genuine multi-objective problem, solved with the full algorithm, fast non-dominated sort, crowding distance, binary tournament selection and elitist $(\mu + \lambda)$ survival, rather than a weighted sum standing in for it. A genome is a vector of k distinct living node indices, so cardinality is fixed by construction. Three objectives are minimized: total energy of one round under the candidate assignment, the negated minimum residual energy among selected heads, and the standard deviation of cluster membership counts. The third was chosen against the more common intra-cluster distance variance, because distance already dominates the first objective through the d^2 and d^4 terms. The first objective calls the engine’s own radio functions, so the search optimizes exactly the cost model it will later be charged against, and selection from the final front is by knee point, which introduces no hand-tuned weights that could act as a back door to calibrating against published results. Evolutionary and swarm formulations have a long history [17], [18]; what is new is not the formulation but that it is charged against the same engine as everything else.

Interval type-2 fuzzy selection scores every living node from residual energy, distance to sink and local density, with Gaussian membership functions of uncertain mean, a 10% footprint of uncertainty, Karnik-Mendel type reduction and midpoint defuzzification [19], [20], [21]. The 27-rule base is generated by score $= 2E + (2 - D) + D_n$ over three-level inputs, giving a table monotone in every input by construction, which is what makes it defensible without tuning; the double weight on residual energy is a designer’s choice and we label it as one. Fuzzy head election has an established line of work [22], [23], [24], and our contribution is not the rule base but its measurement under the same harness as everything else.

C. Generation Three: Learned Selection

SOM [25] trains a Kohonen map with k units over normalized position and residual energy, retrained from scratch on the

same five-round cadence as NSGA-II and the fuzzy system, and the node nearest each converged unit becomes a head.

DQN [26], [27] and GCN [29] are deliberately built to isolate a single difference. Both consume the identical four normalized features, residual energy, distance to sink, local density and rounds since last serving as head, with the GCN importing the DQN’s feature function rather than redefining it. Both use hand-written forward and backward passes verified against central finite differences, and both are trained on seeds 100 to 129, frozen, and evaluated on the disjoint seeds 0 to 29. Only the architecture and the training signal differ.

The DQN is a $4 \rightarrow 16 \rightarrow 2$ multilayer perceptron trained with experience replay, a synchronized target network, Adam [31] and $\gamma = 0.999$. At evaluation, rather than letting each node act on its own greedy action, which would make the head count fluctuate between zero and the entire population, nodes are ranked by the advantage $Q_{CH} - Q_{CH}^-$ and the top k selected. This makes it a learned ranking function rather than a set of independent agents, a real narrowing of the reinforcement-learning framing [32] that we document rather than gloss over.

The GCN is two graph-convolutional layers of width 16 over a symmetrized 6-nearest-neighbor graph, trained self-supervised against a differentiable proxy for energy balance. It is deliberately not trained to imitate a heuristic, since that would make any comparison against that heuristic vacuous, and shares no code path with the DQN’s training loop. Selection is greedy top- k subject to a 15 m minimum separation.

The contrast this buys is the paper’s cleanest: with identical inputs, the DQN has temporal credit assignment and the GCN does not. The GCN can see rotation state through its fourth feature, but nothing in its single-round loss rewards acting on it. Section V-D measures what that costs.

D. Disclosure on the DQN’s Hyperparameters

One protocol’s parameters were revised after observing a failure, and we state it rather than let it be inferred [33]. The initial $\gamma = 0.95$ gave an effective horizon of about twenty rounds against a first-death timescale three orders of magnitude longer, and it diverged. The fix raised γ to 0.999, normalized the return magnitude and extended training from 30 to 100 episodes, leaving the reward function unchanged. A stopping rule was then fixed in advance, recording the result as final regardless of outcome, after which it beat random rotation on all five held-out seeds. Stating the pre-commitment is what makes that number credible, since without it “we fixed the DQN until it worked” and “we fixed a divergence and stopped” are indistinguishable from the outside.

IV. EXPERIMENTAL DESIGN

A. Paired Trials

Every protocol is evaluated over 30 trials in each channel configuration, 600 runs in the headline study, and the trials are paired rather than independent. Trial index i seeds a single random stream generating, in fixed order, the node positions, the initial energies, the full per-link shadowing matrix and

the sensed-value array for all rounds. That order depends only on i and never on which protocol is running, so at a given index all ten configurations face a byte-identical physical world. Protocol-internal randomness comes from a separate stream derived by a keyed hash of run index and protocol name; Python’s built-in hash is salted per process and is never used anywhere in the simulator, since it would silently destroy reproducibility across invocations.

Pairing licenses the statistical procedure below and is substantially stronger than the unpaired mean-and-deviation comparison this literature reports: topology variance is large relative to the differences between protocols, and an unpaired test spends most of its power fighting variance a paired design eliminates by construction. The per-packet channel draws are deliberately not shared, since different protocols make different numbers of draws per round and there is no coherent sense in which two could experience the same coin flips packet by packet. The fairness-critical quantity is the static per-link shadowing, which is identical.

B. The Channel Sensitivity Pair

The entire study is run twice, once with the packet-error model of Section II-C enabled and once with it disabled. This is not a robustness appendix but a decision rule fixed before the results were read:

A conclusion that holds in both channel configurations is a statement about the protocol. A conclusion that changes between them is a statement about transmit power.

Transmit power is a parameter the reference specifications do not fix, and Section II-C showed the packet-error curve is a waterfall whose position determines how severely single-hop protocols are penalized on the sink hop. Choosing it to make a metric discriminating would be precisely the calibration this study forbids, and running both endpoints removes the choice from the argument. Section V-C reports one conclusion the rule retracts.

C. The Node-Count and Field-Area Sweep

A single operating point cannot tell a reader whether a result is about clustering or about one deployment. We repeat the lossy study over a 3×3 grid of node counts $N \in \{50, 100, 150\}$ and square field sides $W \in \{50, 100, 150\}$ m at 15 paired runs per cell, a further 1,350 runs.

The base station scales with the field at $(W/2, 1.5W)$, keeping the deployment geometrically self-similar. This also means field size sets channel severity as well as area: mean node-to-sink distance is 52 m at 50×50 m with no node beyond 120 m, 104 m at 100×100 m with 33% beyond, and 156 m at 150×150 m with 75% beyond. That is a real confound between bigger area and worse channel, and we report mean sink distance per cell so a reader can separate the two rather than engineering the confound away and losing self-similarity instead. The density radius feature scales at $0.25W$, since held fixed at 25 m it would cover 78% of a 50×50 field and 9% of a 150×150 one, silently changing what the feature

means; it is read identically by every protocol, so scaling it is a shared environment choice rather than per-protocol tuning. The DQN and GCN use their frozen $N = 100$, 100×100 weights in every cell with no retraining, which makes the sweep a measurement of out-of-distribution transfer and is why the untrained fuzzy system matters as a control: a protocol with no training at all cannot suffer transfer failure, so if it moves the same way the learned ones do, the cause is the environment. Comparisons inside a cell stay paired; comparisons across cells are not, because a different N means a different number of nodes to seed, and are not tested as if they were.

D. Metrics

Lifetime is reported at first, half and last node death plus the area under the living-node curve, all four together because the three death points do not agree on an ordering. Runs reaching the cap without full depletion are flagged as censored, and a censored last-death figure is a lower bound, never averaged into a mean without the censoring count beside it.

Delivery is reported in two units and the distinction matters. Packets received at the sink measures aggregation depth, not delivery: one chain round delivers a single fused packet where one clustered round delivers roughly five, so on a packet basis the chain protocol appears five times worse while actually delivering more readings. Original readings represented in delivered packets is the cross-protocol comparable quantity, which is why delivery ratio here is reading-based; the packet-based formula reported ten percent on a lossless channel purely because ten source readings became one packet. Suppressed readings count as losses in neither numerator nor denominator, since a reading never injected was never lost, and the cost of suppression appears instead in data yield, readings delivered over readings taken.

Computational cost is reported as wall-clock setup time, counted algorithmic operations and analytical complexity, never merged, because wall-clock reflects implementation quality as much as algorithm design. Timing and memory are measured in separate runs, since allocation tracing inflated setup time by factors between 3.5 and 7.8 depending on how many small objects a protocol allocated, distorting the ranking rather than only the scale.

E. Statistical Procedure and Verification

Because trials are paired, each comparison against a reference reduces to a vector of 30 per-topology differences. The null hypothesis that the difference distribution is symmetric about zero is tested by a sign-flip permutation test [44] over 20,000 random sign vectors, with the standard add-one correction so no p -value is reported as exactly zero; the floor at 30 paired samples is 5×10^{-5} . The procedure was validated against exhaustive enumeration on a reduced sample, giving an exact p of 0.00391 against a sampled 0.00420. Confidence intervals come from 20,000 paired bootstrap resamples [43], and because nine comparisons are made against the reference within each metric and channel, all p -values are corrected by Holm-Bonferroni [42], giving a corrected floor of $p_{\text{Holm}} =$

TABLE III
LIFETIME, LOSSY CHANNEL, 30 PAIRED RUNS (MEAN $\pm$ SAMPLE STD, ROUNDS). FND/HND/LND ARE FIRST, HALF AND LAST NODE DEATH. AUC IS THE AREA UNDER THE LIVING-NODE CURVE, THE CENSORING-ROBUST SURVIVAL MEASURE.

Protocol	Gen.	FND	HND	LND	AUC ($\times 10^3$)
Direct (no clustering)	–	114 $\pm$ 11	1,144 $\pm$ 183	3,202 $\pm$ 78	132.0
LEACH	G1	1,046 $\pm$ 43	1,679 $\pm$ 25	2,191 $\pm$ 81	167.1
PEGASIS	G1	1,588 $\pm$ 331	2,199 $\pm$ 9	2,630 $\pm$ 358	216.4
TEEN	G1	2,208 $\pm$ 164	5,601 $\pm$ 189	>6,999 ^c	535.4
APTEEN	G1	2,144 $\pm$ 139	5,367 $\pm$ 161	>6,999 ^c	518.4
NSGA-II	G2	1,873 $\pm$ 31	1,965 $\pm$ 23	2,697 $\pm$ 256	200.6
Fuzzy T2	G2	1,462 $\pm$ 50	2,006 $\pm$ 28	2,109 $\pm$ 48	191.9
SOM	G3	959 $\pm$ 127	1,998 $\pm$ 32	2,874 $\pm$ 115	196.8
DQN	G3	1,472 $\pm$ 55	1,920 $\pm$ 31	2,325 $\pm$ 59	192.1
GCN	G3	335 $\pm$ 142	1,824 $\pm$ 70	3,185 $\pm$ 142	180.2

^c Censored: the network had not died at the 7000-round cap in any run. Use AUC, not LND, for these two.

0.00045. Permutation was chosen over a paired t -test because it assumes only exchangeability, not normality, and several of these difference distributions are visibly skewed, the GCN's being bimodal; and over Wilcoxon signed-rank because it operates on the actual differences rather than their ranks, so the effect size and the test statistic are the same quantity [45].

Seven gates were required to pass before any number here was read: energy conservation asserted every run to 10^{-9} J against (1); radio energies checked against hand computation on both sides of the crossover; an invariant suite over energy, liveness, death ordering and delivery range; determinism, which all nine protocols satisfy exactly across three repeats per seed; a static audit across eighteen checks confirming no protocol writes to the energy or liveness arrays or chooses its own packet sizes; independent recomputation of one protocol's summary row; and the packet-error curve asserted non-decreasing and strictly between zero and one somewhere in the band of sink distances, without which the channel would be a binary range switch and the lossy results would carry no information.

The headline sweep completes in 80 minutes on sixteen cores and the scale sweep in 222 on fourteen, with figures from Matplotlib [47] and the raw per-round output of every run retained. The central cell of the scale sweep reproduces the first 15 runs of the headline lossy study exactly for all ten protocols, on both first node death and area under the living-node curve, which is the check that the two harnesses are the same simulator.

V. RESULTS

A. What Clustering Actually Buys

The direct-transmission baseline makes the first result concrete. LEACH reaches first node death at 1,046 rounds against the baseline's 114, an improvement of 9.2 times, but the baseline's last node dies at 3,202 rounds against LEACH's 2,191, which is 46% later. Clustering does not extend lifetime; it redistributes it, buying a much later first death at the cost of an earlier last one.

The mechanism is geometric. With the sink at 150 m and a crossover at 87.7 m, 36 of the 100 nodes in the seed-0

TABLE IV

DELIVERY, LOSSY CHANNEL. PACKETS AT THE BASE STATION MEASURE AGGREGATION DEPTH, NOT DELIVERY, AND ARE NOT COMPARABLE ACROSS PROTOCOLS; READINGS DELIVERED AND DATA YIELD ARE. DATA YIELD IS READINGS ARRIVING AT THE SINK DIVIDED BY READINGS TAKEN, SO IT IS THE COLUMN THAT PRICES TEEN AND APTEEN'S SUPPRESSION.

Protocol	Gen.	Packets at BS ($\times 10^3$)	Readings delivered ($\times 10^3$)	PDR (%)	Data yield
Direct (no clustering)	–	128.1	128.1	96.9	0.970
LEACH	G1	11.0	146.6	87.7	0.877
PEGASIS	G1	2.3	182.2	84.2	0.842
TEEN	G1	21.2	58.6	90.8	0.109
APTEEN	G1	21.5	62.6	90.8	0.121
NSGA-II	G2	9.7	179.9	89.6	0.896
Fuzzy T2	G2	9.1	175.4	91.4	0.914
SOM	G3	9.4	175.8	89.3	0.893
DQN	G3	9.0	173.5	90.3	0.903
GCN	G3	8.6	162.8	90.3	0.903

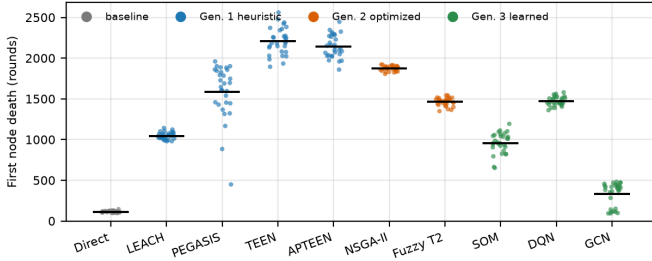


Fig. 2. First node death per trial, lossy channel, all 30 paired trials shown individually rather than as a mean and a deviation. The reactive pair leads, the GCN is visibly bimodal, and PEGASIS carries one topology at 449 rounds against a body near 1,800. A mean and standard deviation describes none of these three distributions correctly.

topology transmit on the free-space branch and are individually cheap to operate. Under direct transmission those nodes are never asked to relay anyone else's traffic and simply outlive everything LEACH produces; under rotation they subsidize the expensive far-field nodes. This is why every lifetime claim here is reported at three death points plus the area under the living-node curve: any single figure silently picks a winner, and the two most commonly reported figures pick opposite ones.

The baseline also gives the clearest illustration of survivorship bias in delivery statistics. Its delivery ratio of 96.9% is the highest of any configuration on the lossy channel, well above LEACH's 87.7%, and it is an artifact, because the distant nodes that lose the most packets die first and the surviving population moves progressively closer to the sink. Measured directly, the baseline's delivery ratio rises from 83.7% over its first hundred rounds to 98.6% over its last hundred. Delivery ratio is never reported here without the living-node curve beside it.

B. Lifetime Across the Three Generations

Table III gives the full lossy-channel results. Against LEACH, with paired permutation testing and Holm correction across the family, every protocol differs significantly. The reactive pair leads on first death by a wide margin, TEEN at 2,208 rounds with $\Delta = +1,163$, followed by NSGA-II at 1,873 ($\Delta = +827$), PEGASIS at 1,588, the DQN at 1,472 and the fuzzy system at 1,462. Two protocols are significantly

worse than LEACH: the self-organizing map at 959 and the GCN at 335.

Three observations survive scrutiny separately from the ranking, and Fig. 2 shows why a table of means would hide all three.

NSGA-II posts the best non-reactive first death and the tightest spread of any of the nine clustering protocols, a standard deviation of 31 rounds against LEACH's 43, the DQN's 55 and PEGASIS's 331, and its advantage holds in both channels. But its convergence is modest and we report that rather than suppress it: over the logged mid-run reclustering the population's best round-energy objective improves by 1.76% and its best minimum-head-energy objective by 3.61%, while only cluster-size balance moves substantially, from 2.99 to 1.85. With five heads drawn from a hundred candidates, random initialization already lands near a good solution, so NSGA-II's advantage should not be attributed to the sophistication of the search.

PEGASIS has by far the largest first-death variance, 331 rounds against a typical 30 to 60, because a greedy chain is sensitive to where the topology places its outliers and one badly positioned node changes the whole cost structure; one trial gives 449 rounds against a body near 1,800. Its half-death standard deviation of 9 rounds is the tightest in the study, because once the chain has been rebuilt a few times its structure becomes topology-insensitive.

The GCN's first death is not merely poor but bimodal. Across the first five seeds it records 448, 394, 418, 112 and 98 rounds, and over all 30 trials eight fall between 94 and 151 while twenty-two fall between 284 and 482. A mean of 335 with a standard deviation of 142 describes neither mode.

C. A Conclusion the Channel Sweep Retracts

Applying the decision rule of Section IV-B, one conclusion does not survive, and it is a load-bearing one. On the lossy channel the fuzzy system and the DQN improve first death over LEACH by 417 and 426 rounds, both at the corrected floor of $p_{\text{Holm}} = 0.00045$. On the ideal channel the same comparisons give 17 and 22 rounds at $p_{\text{Holm}} = 0.119$ and 0.124, not significant. Remove packet loss and both protocols become statistically indistinguishable from LEACH on the metric this literature most often reports. Table V gives both channels side by side.

We measured the mechanism rather than arguing it, and the measurement rules out the explanation we expected. Table VI splits each protocol's energy by accounting category and reports the full distribution of head-to-sink distance, not only its mean.

The obvious reading would be that LEACH's average head sits at 102.1 m, past the 87.7 m crossover and onto the fourth-power branch, while the DQN's sits at 81.1 m. That is true, and it explains part of the difference in raw transmit energy, but it cannot explain the retransmissions. Packet error is 2.6×10^{-8} at 100 m and does not reach 10^{-3} until 120 m, and every protocol in Table VI has a mean head distance below

TABLE V

FIRST-NODE-DEATH DIFFERENCE AGAINST LEACH, IN ROUNDS, UNDER BOTH CHANNEL CONFIGURATIONS. PAIRED SIGN-FLIP PERMUTATION TEST (20 000 PERMUTATIONS) WITH PAIRED-BOOTSTRAP CONFIDENCE INTERVALS, HOLM-CORRECTED ACROSS THE NINE COMPARISONS WITHIN EACH CHANNEL. **FUZZY T2 AND DQN ARE SIGNIFICANT UNDER LOSS AND NOT SIGNIFICANT WITHOUT IT, WHICH IS THE STUDY'S CENTRAL NEGATIVE RESULT.**

Protocol	Lossy channel			Ideal channel		
	Δ FND	95% CI	p_{Holm}	Δ FND	95% CI	p_{Holm}
Direct (no clustering)	-932	[-944, -919]	0.0004	-1,153	[-1,162, -1,144]	0.0004
PEGASIS	+542	[+418, +647]	0.0004	+188	[+78, +293]	0.0090
TEEN	+1,163	[+1,109, +1,217]	0.0004	+2,593	[+2,542, +2,647]	0.0004
APTEEN	+1,099	[+1,055, +1,143]	0.0004	+2,434	[+2,392, +2,480]	0.0004
NSGA-II	+827	[+811, +844]	0.0004	+482	[+471, +493]	0.0004
Fuzzy T2	+417	[+393, +440]	0.0004	+17	[+0, +34]	0.1195
SOM	-87	[-135, -41]	0.0015	-399	[-443, -357]	0.0004
DQN	+426	[+404, +450]	0.0004	+22	[-4, +49]	0.1244
GCN	-710	[-764, -660]	0.0004	-1,140	[-1,193, -1,090]	0.0004

TABLE VI

WHERE THE ENERGY GOES AND WHERE THE HEADS SIT, MEASURED BY THE ENGINE'S OWN ACCOUNTING CATEGORIES (SEED 0, 1100 ROUNDS).

PACKET ERROR IS NEAR ZERO BELOW 120 M, SO THE *mean* HEAD DISTANCE CANNOT EXPLAIN THE RETRY GAP, BECAUSE EVERY MEAN IS BELOW THAT. THE TAIL CAN: ACROSS THE SIX SINGLE-HOP PROTOCOLS, THE SHARE OF HEAD-ROUNDS BEYOND 120 M PREDICTS RETRY ENERGY SHARE WITH $r = 0.95$. PEGASIS IS THE EXCEPTION THAT PROVES THE RULE: ITS LEADER ROTATES UNIFORMLY AND 30% OF LEADER-ROUNDS SIT BEYOND 120 M, YET IT RETRIES BARELY AT ALL, BECAUSE ONLY THE LEADER EVER FACES THE SINK.

Protocol	Retry (%)	Control (%)	Head-to-sink (m)		Heads >120 m (% of rounds)
			mean	p90	
LEACH	6.39	12.73	102.1	143.5	29.9
PEGASIS	1.39	4.38	102.2	143.5	30.0
SOM	5.28	2.38	102.8	136.4	31.1
NSGA-II	3.57	2.46	94.2	127.5	17.7
Fuzzy T2	3.65	2.33	85.6	122.1	11.5
DQN	2.31	3.95	81.1	111.2	5.1
GCN	2.25	3.85	77.1	104.6	3.2

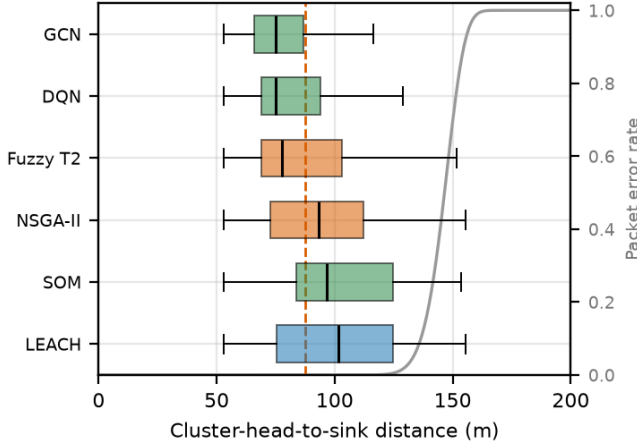


Fig. 3. Distribution of head-to-sink distance per protocol, drawn over the packet-error curve. Every protocol's *mean* falls in the flat region where packet error is effectively zero, so the mean cannot drive the retry gap. The distributions differ in their upper tails, and only the tails reach the waterfall.

120 m. A protocol whose heads all sat at the mean would retry essentially never.

What separates them is the upper tail, shown in Fig. 3. LEACH spends 29.9% of its head-rounds beyond 120 m and 6.39% of its budget on retransmissions; the DQN spends 5.1% of head-rounds beyond 120 m and 2.31% on retransmissions. Across the six single-hop protocols the tail share correlates with retry energy share at $r = 0.95$. The mean distance correlates almost as strongly, at $r = 0.93$, because the two are collinear; the reason to prefer the tail is not a better fit but that only the tail lies in the region where packets actually fail.

PEGASIS fixes the scope of the claim and we state it rather than exclude it quietly. Its leader rotates uniformly over chain positions, so 30.0% of leader-rounds sit beyond 120 m, the same as LEACH, yet it spends only 1.39% on retries, and including it drops the correlation from 0.95 to 0.50. The reason is structural: in a chain exactly one node faces the sink per round while every other transmission is a short hop, so the tail is paid once per round instead of five times. The claim is therefore specific to single-hop clustered protocols, where every head pays the sink hop.

The honest claim is narrower, and more interesting, than “the soft-computing and learned protocols cluster better”. Their demonstrable skill is keeping the far tail of the head-to-sink hop out of the error waterfall, and that skill pays only in proportion to how lossy the hop is. The same table shows LEACH's compensating strength: it consumes 20% more total energy than the DQN yet matches its first death on a clean channel, because its epoch rule rotates head duty perfectly.

The retraction is specific to first death and we do not overstate it. On area under the living-node curve and on readings delivered, all eighteen comparisons against LEACH are significant at the corrected floor in both channels, in the same direction and at similar magnitude; the fuzzy system gains 24,793 node-rounds on the lossy channel and 22,120 on the ideal. Across all forty protocol-metric rankings only four ranks move between channels, and both moving pairs are near-ties between adjacent protocols. The ordering is stable; what is not stable is the significance of specific first-death gaps. First death is a single order statistic and the most fragile of the four

TABLE VII

HEAD-DUTY CONCENTRATION OVER 1100 ROUNDS (SEED 0), IN ROUNDS SERVED AS CLUSTER HEAD PER NODE. LEACH'S 55 IS EXACT RATHER THAN APPROXIMATE: ITS EPOCH MECHANISM ELECTS EVERY NODE EXACTLY ONCE PER 20 ROUNDS. GCN'S BUSIEST NODE SERVES 13.6 TIMES AS OFTEN, WHICH IS THE DIRECT CAUSE OF ITS EARLY FIRST NODE DEATH (TABLE III): IT DOES NOT CHOOSE BADLY, IT CHOOSES THE SAME GOOD NODE UNTIL THAT NODE DIES.

Protocol	Busiest node	Median	Never served	Gini
LEACH	55	55	0	0.001
PEGASIS	11	11	0	0.000
NSGA-II	95	60	0	0.207
Fuzzy T2	360	40	4	0.416
SOM	175	50	10	0.393
DQN	223	38	16	0.476
GCN	750	16	13	0.700

lifetime measures.

D. What Temporal Credit Assignment Buys

The DQN and the GCN consume identical inputs and differ only in architecture and training signal, so the gap between them isolates one variable. Over the 30 paired trials the DQN's first death ranges from 1,361 to 1,580, a spread of 219 rounds with a standard deviation of 55; the GCN's ranges from 94 to 482, a spread of 388 with a standard deviation of 142, and it is bimodal rather than merely wide.

Table VII measures why over 1,100 rounds on seed 0. The GCN's busiest node serves as cluster head for 750 of those rounds against LEACH's 55, a factor of 13.6, and its Gini coefficient over head duty is 0.700 against LEACH's 0.001; thirteen of its nodes never serve at all and its median node serves 16 rounds. It does not choose badly. It repeatedly chooses the same genuinely good node until that node dies, which is exactly the failure a single-round objective cannot see. The DQN, with the same four features including rounds since last serving as head, concentrates far less, at 223 rounds on its busiest node and a Gini of 0.476.

This is the paper's cleanest controlled contrast. The GCN can see rotation state in its input; what it lacks is any term in its loss that rewards deferring cost to extend lifetime, and the cost of that omission is a first death 3.1 times earlier than LEACH's. It is also the only protocol to lose to LEACH on first death by a wide margin in both channels.

E. Delivery, Cost and the Reactive Tradeoff

TEEN and APTEEN dominate every lifetime metric and fail to reach full depletion within the 7,000-round cap in all 30 trials, so their last-death figures are lower bounds and their area under the living-node curve is 3.2 times LEACH's. This is not free. Their data yield is 0.109 and 0.121 against roughly 0.9 for the proactive protocols, and they deliver 58,551 and 62,579 readings against LEACH's 146,616. They survive by not reporting, and their reporting rate, 0.15 and 0.17 of rounds, is what makes the lifetime figure interpretable; reported without it, their lifetime is not a result about clustering at all.

Table IV also shows why packets at the base station must not be compared across protocols. PEGASIS delivers 182,179

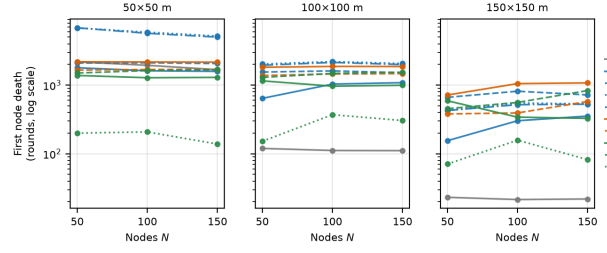


Fig. 4. First node death across the nine cells, log scale, color by generation. Moving between field sizes shifts every protocol by roughly an order of magnitude; moving between node counts within a field size barely moves anything. The reactive pair reverses rank between the smallest and largest fields.

readings in 2,272 packets, an aggregation ratio of 80, while the direct baseline delivers 128,075 readings in as many packets: on packet count PEGASIS looks like the worst protocol in the study and on readings it is the best.

The three notions of computational cost disagree, which is the argument for keeping them separate. NSGA-II costs 102.9 ms per round of setup against LEACH's 0.82 ms and 86,348 counted operations against 329. But the orderings differ: the DQN performs 8,210 operations per round in 0.52 ms while the fuzzy system performs 2,301 in 5.14 ms, so the DQN does 3.6 times the algorithmic work in a tenth of the time, its work being a dense matrix product against an iterative type-reduction loop, and reporting either number alone would invert the comparison. Latency separates the protocols structurally, the clustered protocols falling between 387 and 419 ms mean against PEGASIS at 1,045 ms, because chain depth dominates.

F. Node Count, Field Area, and the Regime Boundary

The headline study measures one deployment. The 3×3 sweep asks which of its conclusions are about clustering and which are about that deployment, and Table VIII gives first node death in all nine cells.

Area dominates node count by roughly five to one. Averaged over all ten configurations, tripling the node count changes first node death by a factor of 0.88 to 1.26, while tripling the field side changes it by a factor of 5 to 7. Fig. 4 shows the separation directly.

The driver is distance to the sink, not density. Ordering the cells by node density produces no trend; they track mean sink distance instead. The two cells closest in density, 50 and 44 nodes per hectare, differ by 1.7 times for NSGA-II, 2.1 times for LEACH and 5.5 times for direct transmission, because one has its sink 104 m away and the other 156 m. Density spans a factor of 27 across the grid and explains almost none of the variation. The grid was not designed to hold density constant, so this is a near-match rather than a controlled contrast; the effect sizes differ by enough that the conclusion is not sensitive to the gap, but the design does not license a stronger statement.

Clustering is harmful when the sink is close. In the 50×50 m field, where mean node-to-sink distance is 52 m and no node lies beyond 120 m, LEACH's first node death relative to no clustering at all is a factor of 0.82, 0.83 and 0.95 at

TABLE VIII
FIRST NODE DEATH (ROUNDS) ACROSS NODE COUNTS AND FIELD AREAS, LOSSY CHANNEL, 15 PAIRED RUNS PER CELL.

Protocol	Gen.	50×50 m			100×100 m			150×150 m		
		N=50	N=100	N=150	N=50	N=100	N=150	N=50	N=100	N=150
Direct (no clustering)	–	2,178	1,943	1,677	120	112	111	23	22	22
LEACH	G1	1,791	1,617	1,586	641	1,034	1,085	155	303	354
PEGASIS	G1	2,097	2,114	2,062	1,550	1,610	1,517	663	815	720
TEEN	G1	6,808	5,899	5,186	2,028	2,201	2,064	425	550	532
APTEEN	G1	6,831	5,684	5,002	1,947	2,141	1,992	427	519	527
NSGA-II	G2	2,184	2,173	2,160	1,818	1,872	1,865	717	1,052	1,077
Fuzzy T2	G2	1,656	1,685	1,698	1,379	1,454	1,470	381	393	577
SOM	G3	1,382	1,280	1,296	1,153	965	994	591	342	328
DQN	G3	1,500	1,628	1,677	1,296	1,478	1,537	454	559	833
GCN	G3	200	208	138	152	371	305	71	158	82

The base station scales with the field at $(W/2, 1.5W)$, so field size sets channel severity as well as area. Node-to-sink distance by field size. 50×50 m: mean 52 m, 0% of nodes beyond 120 m; 100×100 m: mean 104 m, 33% of nodes beyond 120 m; 150×150 m: mean 156 m, 75% of nodes beyond 120 m. At 50 × 50 m no link reaches the error waterfall at all, which is why the advantage of the optimized and learned protocols over LEACH collapses there.

$N = 50$, 100 and 150, and the no-clustering baseline outlives LEACH in all three cells on area under the living-node curve as well as on first death. At 100 × 100 m the same ratios are 5.4, 9.3 and 9.8, and at 150 × 150 m they are 6.7, 13.8 and 16.1. The opening result of this section is therefore not a statement about clustering as such but about an operating regime: clustering pays once the sink hop is expensive enough to be worth aggregating away, and not before.

The channel retraction reproduces from an independent direction. In the 50 × 50 m field the DQN’s advantage over LEACH falls to +10 rounds at $N = 100$ and to −291 at $N = 50$, against +444 at 100 × 100 m; the fuzzy system falls to +68 and −135 against +419. This is the boundary of Section V-C reached by moving the sink close enough that no link reaches the waterfall, rather than by removing packet loss. The fuzzy system makes the attribution safe: it has no training of any kind, so its collapse cannot be transfer failure and must be a property of the environment. Since the DQN behaves the same way, the DQN’s collapse is too.

Rankings are partly stable and the exceptions are informative. NSGA-II holds ranks 1 to 3 in every cell, takes rank 1 in all three 150 × 150 m cells, and beats LEACH in all nine by margins from +394 to +1,177 rounds, while the GCN holds ranks 9 to 10 everywhere. Against those, the DQN’s rank improves monotonically as the field grows, and TEEN and APTEEN reverse outright, from ranks 1 and 2 in the two smaller fields to ranks 4 to 6 at 150 × 150 m. Rankings by area under the living-node curve are far more stable, with every protocol except the direct baseline moving at most four places.

The frozen policies do not collapse off their training distribution. Carrying $N = 100$, 100 × 100 m weights into all nine cells, neither learned protocol fails catastrophically, and the GCN’s deficit against LEACH shrinks as the field grows, from between −1,409 and −1,591 rounds in the smallest field to between −84 and −272 in the largest, though it remains worse than LEACH everywhere. This is weaker evidence than a retrained comparison would give and is not a claim of scale invariance; it is a claim that the reported advantage is not an artifact of evaluating these policies only where they were

trained.

Two of our five pre-registered expectations were wrong and we report both. We expected the generational gap to widen at 150 × 150 m; it narrows there as well as at 50 × 50 m, so the gap is largest in the middle cell. And we expected node density to matter; it does not.

VI. THREATS TO VALIDITY

Every result in Section V depends on choices a reader cannot see in the numbers. This section states them, ordered by how far each could move a conclusion, and names the one that could move the most first.

A. The Control-Traffic Subsidy

Centralized protocols are charged nothing for the uplink that informs the sink of node state. This is the largest single confound in the study and it favors five of the ten configurations. The justification is quantitative: a separate 200-bit status report per node per round, over a sink distance above 100 m and therefore on the fourth-power branch, costs roughly 3.6×10^{-5} J per node per round, about a quarter of the entire 1 J budget across the round cap. Charging it would sink all five centralized protocols for a reason unconnected to whether their clustering decisions are any good, and the piggyback assumption is standard in this literature. But a defensible modeling choice is still a choice, and its magnitude belongs in the results rather than in a footnote.

Measured over the full sweep, LEACH spends 12.73% of its consumed energy on control traffic against 2.33% for the fuzzy system, 2.46% for NSGA-II, 2.38% for the map, 3.95% for the DQN and 3.85% for the GCN, so the subsidy is worth roughly nine to ten percentage points of the total budget. In packet terms the gap is starker, 171,644 control packets for LEACH against 9,657 to 12,216 for the centralized five, but that overstates the effect, since control packets are a fifth the size of data packets and most are short-range.

Two consequences follow. A claim of the form “NSGA-II extends first death by 80% over LEACH” conflates clustering quality with an accounting convention and should not be

made without qualification. And the reliable comparisons here are the within-class ones, centralized against centralized and distributed against distributed, with the chain protocol comparing cleanly to neither. The strongest results in Section V, the channel-conditional retraction and the DQN-versus-GCN rotation contrast, are both within-class and unaffected. The clean resolution is an ablation that charges the uplink honestly and re-runs the sweep, which would convert this caveat into a measurement. It has not been run.

B. Transmit Power Is Bracketed, Not Swept

Section V-C treats the channel comparison as a result, which it is, but two configurations bracket a continuum without tracing it. A transmit power between the endpoints, or above the lossy one, would produce intermediate behavior this study does not measure, and a three-point or four-point sweep would show where the fuzzy system's and the DQN's advantage actually crosses the significance threshold. That sweep has not been run. The scale sweep partially compensates, since moving the sink changes how much of the population sits in the waterfall, but it is not a substitute: field size changes area and channel severity together, so it locates the boundary without isolating the parameter. The path-loss exponent, shadowing standard deviation, noise floor and retry budget are equally unspecified in the source material and equally consequential. We used conventional values for 2.4 GHz outdoor short-range links [34], [36] and published the resulting packet-error curve so the choice is auditable, but we did not vary them.

C. The Pre-Training Asymmetry

The DQN and the GCN arrive pre-trained; the other seven do not. Their reported setup time and operation counts are inference costs only, and training is real, paid once and reported separately, so it must not be read as free. The seven non-learned protocols receive no equivalent offline preparation, which makes this a fair comparison of a deployed system and not an apples-to-apples algorithmic comparison; both statements are true and which matters depends on the question being asked.

Generalization is now measured rather than assumed, but only in one direction. The evaluation seeds are disjoint from the training seeds, so the headline result is not memorization, and the scale sweep carries the frozen weights into eight cells the policies never saw without either collapsing. What that does not establish is how these policies would compare against retrained versions of themselves, or how they behave under a sink placement not geometrically similar to the training one, since every cell keeps the base station at $(W/2, 1.5W)$.

D. Modeling Simplifications

The radio model accounts for transmit, receive and fusion energy and nothing else: no idle listening, no sleep-wake transitions, no radio startup, no processor energy, and a linear reservoir with no rate-capacity or recovery effect. Idle listening frequently dominates real deployments and including it would

compress the differences between all ten configurations toward zero.

Fusion is ideal: a head receiving any number of member packets emits exactly one, so compression is lossless and free in bits. This systematically favors high-fan-in structures and PEGASIS most of all, whose aggregation ratio of 80 readings per delivered packet is exactly the quantity the assumption idealizes, and it shapes the chain-versus-cluster comparison more than any other single simplification.

There is no MAC layer, so scheduling is perfect and collision-free and packet loss comes only from distance-dependent error [37], [38]. Adding interference would penalize dense-cluster protocols relative to the chain and would make the number of simultaneously active heads a variable this study cannot see, and latency inherits the same limitation, so its absolute figures are lower bounds whose ordering is meaningful and whose magnitudes are not. Everything is static, and the head-to-sink hop is single-hop for nine of ten configurations, matching each protocol's original specification and leaving the chain as the sole multi-hop contrast. Sensor data is a mean-reverting process rather than a real trace, so the reactive protocols' results are conditional on that model. Finally, every protocol targets the same nominal head count but realizes it differently, and LEACH's realized count is deliberately left stochastic, so part of every observed difference is attributable to realized cluster count rather than to selection logic.

E. Scope and Non-Calibration

This study establishes results at one energy budget and one traffic model. Network scale is swept over three node counts and three field areas, but within a single family of geometrically similar deployments, and three protocols have complexity superlinear in node count. Node energy is homogeneous, every node senses every round with no duty cycling or query workload, and there is no hardware testbed and no measured trace; every number is simulation output under the assumptions above.

No protocol was tuned to reproduce a published figure. Only protocol descriptions, the radio model and simulation parameters were taken from prior work, and its numeric results were discarded before implementation began, so disagreement with published tables is expected and is not evidence of error here. The one place this commitment was tested is documented in Section III-D, where the DQN's discount factor and training length were revised after an observed divergence, after which a stopping rule was fixed in advance and the result recorded as final regardless of outcome. We state this because the alternative, presenting the final hyperparameters as if they had been the first, is the ordinary way such revisions disappear from a methods section [7], [33].

VII. CONCLUSION

We built a single simulation engine in which nine clustering protocols spanning twenty-five years of design practice execute under provably identical physical conditions, and

evaluated them over 600 runs across two channel configurations and a further 1,350 across nine deployment scales. The engine owns energy accounting, packet sizing, fusion charges, the channel and reading accounting; a protocol returns only a cluster structure. Fairness is therefore a property of the construction rather than of the authors' care, and it is verified by static audit rather than asserted.

Clustering does not extend network lifetime so much as redistribute it. LEACH delays first node death by a factor of 9.2 over direct transmission while bringing last death forward by nearly a third, because about a third of the nodes are close enough to the sink to transmit on the free-space branch and are penalized rather than helped by rotation. No single lifetime figure summarizes this, and the two most commonly reported figures order the protocols in opposite directions.

The advantage of the soft-computing and learned protocols over LEACH in first node death is conditional on the channel: under packet loss the fuzzy system and the Q-network improve on LEACH by 417 and 426 rounds, and with loss removed the same comparisons give 17 and 22 rounds and lose significance. Measurement of the energy split shows why, and not in the way the mean distances suggest. Every protocol's mean head-to-sink distance sits below 120 m, where packet error is still effectively zero, so the mean cannot be the cause; the tails differ instead, LEACH putting 29.9% of its head-rounds beyond 120 m at 6.39% of its budget in retransmissions against the Q-network's 5.1% and 2.31%. The demonstrable skill of the newer protocols is keeping the far tail of the sink hop out of the error waterfall, and it pays in proportion to how lossy that hop is. Their advantage in overall survival and in readings delivered is not conditional and holds in both configurations at the corrected floor.

The same boundary appears independently when the deployment scales. Tripling the node count moves first node death by a factor of 0.88 to 1.26 while tripling the field side moves it by 5 to 7, and cells of nearly equal density differ by up to 5.5 times, so distance to the sink is the variable and density is not. In a 50×50 m field, where nothing reaches the waterfall, the learned and soft-computing advantages collapse exactly as they do when packet loss is switched off, and clustering itself becomes counterproductive: the no-clustering baseline outlives LEACH in all three of those cells.

Temporal credit assignment is worth measuring separately from representational capacity. The Q-network and the graph network consume identical features and differ only in architecture and training signal, yet the Q-network's first-death distribution spans 219 rounds while the graph network's spans 388, is bimodal, and concentrates head duty on one node 13.6 times as often as LEACH does while 13 nodes never serve at all. A single-round objective cannot represent deferring cost to extend lifetime even when rotation state is visible in its inputs. Reactive reporting, meanwhile, buys lifetime at a price that must be reported alongside it: TEEN and APTEEN survive past the round cap in every trial with an area under the living-node curve more than three times LEACH's, at a data yield of 0.11 to 0.12 against roughly 0.9 for the proactive protocols.

The methodological point generalizes beyond these nine protocols. Running the entire study at both endpoints of an unspecified parameter, and committing in advance to report only what survives both, retracted a comparison that either endpoint alone would have supported. Where such a parameter sits near a nonlinearity, as transmit power does relative to the packet-error waterfall, the choice can decide the ranking. The largest confound remaining is the accounting subsidy granted to centralized protocols, and an ablation charging the uplink honestly would convert it from a caveat into a measurement. All configuration constants, per-round outputs and verification artifacts are released with the code.

REFERENCES

- [1] I. F. Akyildiz, W. Su, Y. Sankarasubramaniam, and E. Cayirci, "Wireless Sensor Networks: A Survey," *Computer Networks*, vol. 38, no. 4, pp. 393–422, 2002.
- [2] A. A. Abbasi and M. Younis, "A Survey on Clustering Algorithms for Wireless Sensor Networks," *Computer Communications*, vol. 30, no. 14–15, pp. 2826–2841, 2007.
- [3] M. A. Alsheikh, S. Lin, D. Niyato, and H.-P. Tan, "Machine Learning in Wireless Sensor Networks: Algorithms, Strategies, and Applications," *IEEE Communications Surveys and Tutorials*, vol. 16, no. 4, pp. 1996–2018, 2014.
- [4] S. Kurkowski, T. Camp, and M. Colagrosso, "MANET Simulation Studies: The Incredibles," *ACM SIGMOBILE Mobile Computing and Communications Review*, vol. 9, no. 4, pp. 50–61, 2005.
- [5] K. Pawlikowski, H.-D. J. Jeong, and J.-S. R. Lee, "On Credibility of Simulation Studies of Telecommunication Networks," *IEEE Communications Magazine*, vol. 40, no. 1, pp. 132–139, 2002.
- [6] T. R. Andel and A. Yasinsac, "On the Credibility of MANET Simulations," *Computer*, vol. 39, no. 7, pp. 48–54, 2006.
- [7] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, E. Fox, H. Larochelle, *et al.*, "Improving Reproducibility in Machine Learning Research: A Report from the NeurIPS 2019 Reproducibility Program," *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021.
- [8] W. R. Heinzelman, A. Chandrakasan, and H. Balakrishnan, "Energy-Efficient Communication Protocol for Wireless Microsensor Networks," in *Proc. 33rd Annual Hawaii International Conference on System Sciences (HICSS)*, pp. 1–10, 2000.
- [9] W. B. Heinzelman, A. P. Chandrakasan, and H. Balakrishnan, "An Application-Specific Protocol Architecture for Wireless Microsensor Networks," *IEEE Transactions on Wireless Communications*, vol. 1, no. 4, pp. 660–670, 2002.
- [10] S. Lindsey and C. S. Raghavendra, "PEGASIS: Power-Efficient Gathering in Sensor Information Systems," in *Proc. IEEE Aerospace Conference*, vol. 3, pp. 1125–1130, 2002.
- [11] A. Manjeshwar and D. P. Agrawal, "TEEN: A Routing Protocol for Enhanced Efficiency in Wireless Sensor Networks," in *Proc. 15th International Parallel and Distributed Processing Symposium (IPDPS) Workshops*, pp. 2009–2015, 2001.
- [12] A. Manjeshwar and D. P. Agrawal, "APTEEN: A Hybrid Protocol for Efficient Routing and Comprehensive Information Retrieval in Wireless Sensor Networks," in *Proc. 16th International Parallel and Distributed Processing Symposium (IPDPS)*, pp. 195–202, 2002.
- [13] O. Younis and S. Fahmy, "HEED: A Hybrid, Energy-Efficient, Distributed Clustering Approach for Ad Hoc Sensor Networks," *IEEE Transactions on Mobile Computing*, vol. 3, no. 4, pp. 366–379, 2004.
- [14] G. Smaragdakis, I. Matta, and A. Bestavros, "SEP: A Stable Election Protocol for Clustered Heterogeneous Wireless Sensor Networks," in *Proc. 2nd International Workshop on Sensor and Actor Network Protocols and Applications (SANPA)*, 2004.
- [15] L. Qing, Q. Zhu, and M. Wang, "Design of a Distributed Energy-Efficient Clustering Algorithm for Heterogeneous Wireless Sensor Networks," *Computer Communications*, vol. 29, no. 12, pp. 2230–2237, 2006.
- [16] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II," *IEEE Transactions on Evolutionary Computation*, vol. 6, no. 2, pp. 182–197, 2002.

- [17] K. P. Ferentinos and T. A. Tsiligiridis, "Adaptive Design Optimization of Wireless Sensor Networks Using Genetic Algorithms," *Computer Networks*, vol. 51, no. 4, pp. 1031–1051, 2007.
- [18] R. V. Kulkarni and G. K. Venayagamoorthy, "Particle Swarm Optimization in Wireless-Sensor Networks: A Brief Survey," *IEEE Transactions on Systems, Man, and Cybernetics, Part C*, vol. 41, no. 2, pp. 262–267, 2011.
- [19] J. M. Mendel and R. I. B. John, "Type-2 Fuzzy Sets Made Simple," *IEEE Transactions on Fuzzy Systems*, vol. 10, no. 2, pp. 117–127, 2002.
- [20] N. N. Karnik and J. M. Mendel, "Centroid of a Type-2 Fuzzy Set," *Information Sciences*, vol. 132, no. 1–4, pp. 195–220, 2001.
- [21] D. Wu and J. M. Mendel, "Enhanced Karnik-Mendel Algorithms," *IEEE Transactions on Fuzzy Systems*, vol. 17, no. 4, pp. 923–934, 2009.
- [22] I. Gupta, D. Riordan, and S. Sampalli, "Cluster-Head Election Using Fuzzy Logic for Wireless Sensor Networks," in *Proc. 3rd Annual Communication Networks and Services Research Conference (CNSR)*, pp. 255–260, 2005.
- [23] J.-M. Kim, S.-H. Park, Y.-J. Han, and T.-M. Chung, "CHEF: Cluster Head Election Mechanism Using Fuzzy Logic in Wireless Sensor Networks," in *Proc. 10th International Conference on Advanced Communication Technology (ICACT)*, 2008.
- [24] H. Bagci and A. Yazici, "An Energy Aware Fuzzy Approach to Unequal Clustering in Wireless Sensor Networks," *Applied Soft Computing*, vol. 13, no. 4, pp. 1741–1749, 2013.
- [25] T. Kohonen, "The Self-Organizing Map," *Proceedings of the IEEE*, vol. 78, no. 9, pp. 1464–1480, 1990.
- [26] C. J. C. H. Watkins and P. Dayan, "Q-Learning," *Machine Learning*, vol. 8, no. 3–4, pp. 279–292, 1992.
- [27] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, G. V. Bellemare, and others, "Human-Level Control Through Deep Reinforcement Learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.
- [28] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.
- [29] T. N. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in *Proc. International Conference on Learning Representations (ICLR)*, 2017.
- [30] Z. Wu, S. Pan, F. Chen, G. Long, C. Zhang, and P. S. Yu, "A Comprehensive Survey on Graph Neural Networks," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 1, pp. 4–24, 2021.
- [31] D. P. Kingma and J. Ba, "Adam: A Method for Stochastic Optimization," in *Proc. International Conference on Learning Representations (ICLR)*, 2015.
- [32] N. C. Luong, D. T. Hoang, S. Gong, D. Niyato, P. Wang, Y.-C. Liang, and D. I. Kim, "Applications of Deep Reinforcement Learning in Communications and Networking: A Survey," *IEEE Communications Surveys and Tutorials*, vol. 21, no. 4, pp. 3133–3174, 2019.
- [33] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep Reinforcement Learning That Matters," in *Proc. 32nd AAAI Conference on Artificial Intelligence*, pp. 3207–3214, 2018.
- [34] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002.
- [35] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York, NY, USA: McGraw-Hill, 2008.
- [36] IEEE, *IEEE Standard for Low-Rate Wireless Networks*, IEEE Std. 802.15.4-2020, 2020.
- [37] J. Zhao and R. Govindan, "Understanding Packet Delivery Performance in Dense Wireless Sensor Networks," in *Proc. 1st International Conference on Embedded Networked Sensor Systems (SenSys)*, pp. 1–13, 2003.
- [38] A. Woo, T. Tong, and D. Culler, "Taming the Underlying Challenges of Reliable Multihop Routing in Sensor Networks," in *Proc. 1st International Conference on Embedded Networked Sensor Systems (SenSys)*, pp. 14–27, 2003.
- [39] W. Ye, J. Heidemann, and D. Estrin, "An Energy-Efficient MAC Protocol for Wireless Sensor Networks," in *Proc. IEEE INFOCOM*, pp. 1567–1576, 2002.
- [40] J. Polastre, J. Hill, and D. Culler, "Versatile Low Power Media Access for Wireless Sensor Networks," in *Proc. 2nd International Conference on Embedded Networked Sensor Systems (SenSys)*, pp. 95–107, 2004.
- [41] G. E. Uhlenbeck and L. Ornstein, "On the Theory of the Brownian Motion," *Physical Review*, vol. 36, no. 5, pp. 823–841, 1930.
- [42] S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [43] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York, NY, USA: Chapman and Hall/CRC, 1993.
- [44] P. I. Good, *Permutation, Parametric, and Bootstrap Tests of Hypotheses*, 3rd ed. New York, NY, USA: Springer, 2005.
- [45] J. Demsar, "Statistical Comparisons of Classifiers over Multiple Data Sets," *Journal of Machine Learning Research*, vol. 7, pp. 1–30, 2006.
- [46] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, *et al.*, "Array Programming with NumPy," *Nature*, vol. 585, no. 7825, pp. 357–362, 2020.
- [47] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," *Computing in Science and Engineering*, vol. 9, no. 3, pp. 90–95, 2007.